A Localized, Linear Learning Error Mitigation Approach Using QPD for NISQ Devices

Presenter: Tanvir Ahmed

Degree: Bachelor of Science in Computer Science and Engineering

Supervisor: Dr. Rahat Hossain Faisal

University: University of Barisal

6 January 2026

Presentation Outline

- Motivation
- Background (NISQ & QEM)
- Problem Statement & Goals
- Methodology (PDR-QPD Framework)
- Experimental Setup
- Results (Simulated & Real Hardware)

Motivation: Why this problem?

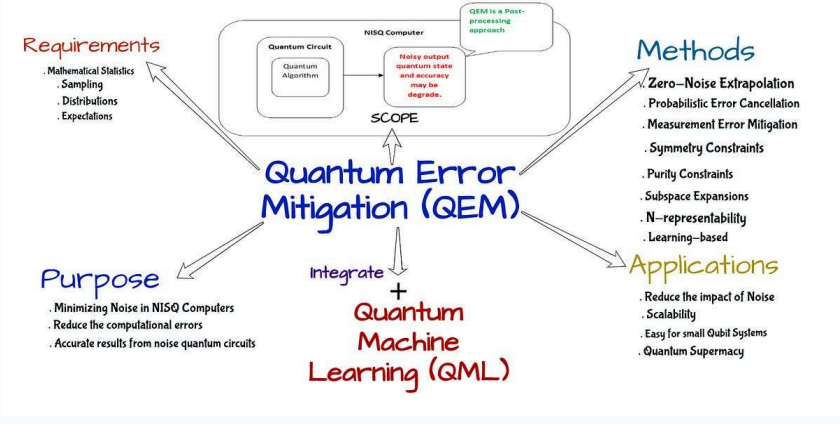
- **NISQ Reality:** Current quantum devices are noisy; deep circuits fail due to decoherence and gate errors.
- **Scalability Issues:** Existing mitigation methods like ZNE (Zero-Noise Extrapolation) require increasing circuit depth, which scales poorly.
- **The Need:** We need a resource-efficient mitigation strategy specifically designed for distributed/cut circuits (QPD).



Noise + Cuts = Need for Correction

Background: NISQ & QEM

- **NISQ (Noisy Intermediate-Scale Quantum):** 50-100 qubits, no fault tolerance yet.
- **Why Errors Matter:** Noise (decoherence, crosstalk) corrupts computation results.
- **Common QEM Methods:**
  - ZNE: Extrapolates to zero noise (Drawback: amplifies variance).
  - PEC: Probabilistic Cancellation (Drawback: high sampling cost).
  - Learning-based: Neural Networks (Drawback: high training cost).



Overview of Quantum Error Mitigation Strategies

Problem Statement & Goals

The Problem

Running large quantum circuits via cutting (QPD) enables execution on smaller devices, but the reconstructed fragments still suffer from inherent hardware noise.

Research Objective

To develop a **localized**, **low-sample** error mitigation framework specifically for QPD fragments that does not increase circuit depth.

High-Level Approach: PDR-QPD Pipeline

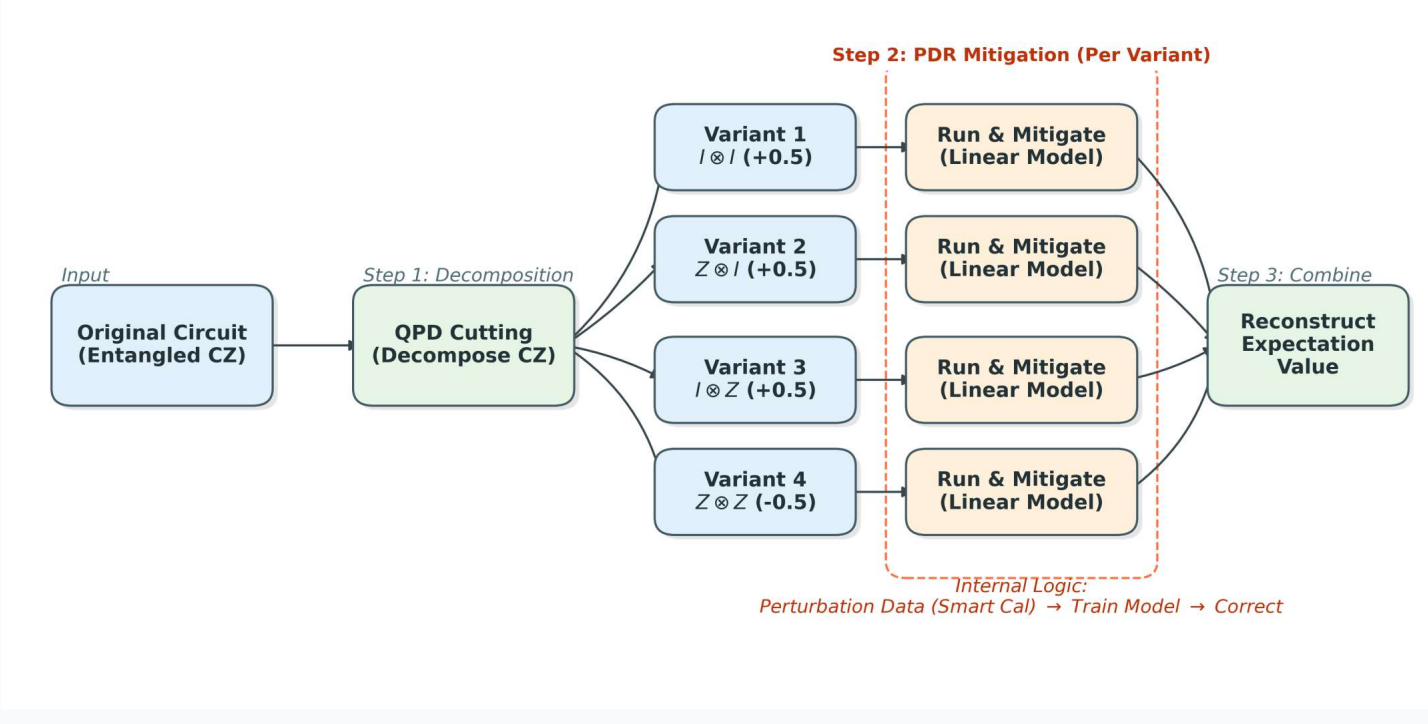


Figure 3.1: *The PDR-QPD Pipeline: Decomposition → Calibration (Perturbation) → Local Regression → Reconstruction.*

Method: Quasi-Probability Decomposition (QPD)

Decomposition of CZ Gate:

Non-local gates are broken into local tensor products to fit hardware constraints.

$$\mathrm{CZ} = \frac{1}{2}(I \otimes I) + \frac{1}{2}(Z \otimes I) + \frac{1}{2}(I \otimes Z) - \frac{1}{2}(Z \otimes Z)$$

4 Variants:

- Variant 1: I ⊗ I
- Variant 2: Z ⊗ I
- Variant 3: I ⊗ Z
- Variant 4: Z ⊗ Z

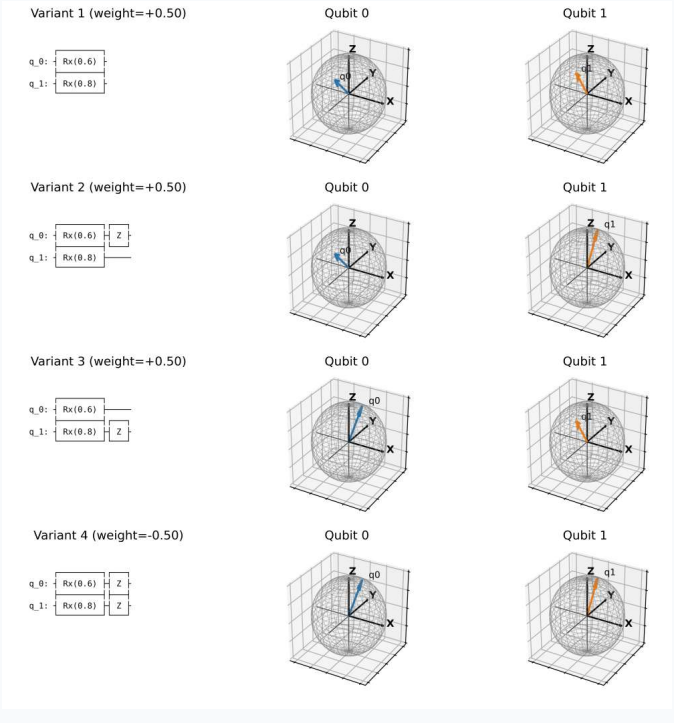


Figure 3.2: *Local circuit variants derived from QPD.*

Calibration: Structure-Aware Perturbation

What we do:

- Apply small perturbations ($\delta\theta$) to rotation angles.
- Range: **[-0.2, 0.2]**.
- Generate **K** calibration circuits.

Why?

- Preserves circuit structure (same equivalence class).
- Avoids depth blow-up (unlike ZNE).
- Captures local noise profile accurately.

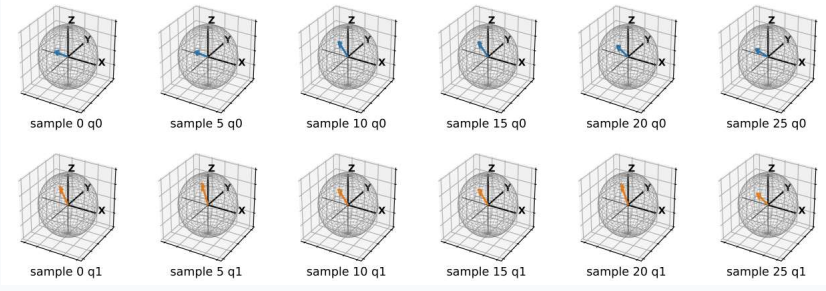


Figure 3.3: *Structure-aware perturbation samples on the Bloch sphere.*

Method: Local Linear Regression (PDR)

- **Training Data:** Pairs of (E_{noisy}, E_{ideal}).
- **Model:** Fit a linear model per QPD variant.
- **Estimator:** Theil-Sen (robust to outliers) + OLS fallback.
- **Reconstruction:**

$$\langle O \rangle_{\text{mit}} = \sum_i w_i \cdot f_i(E^{(i)}_{\text{raw}})$$

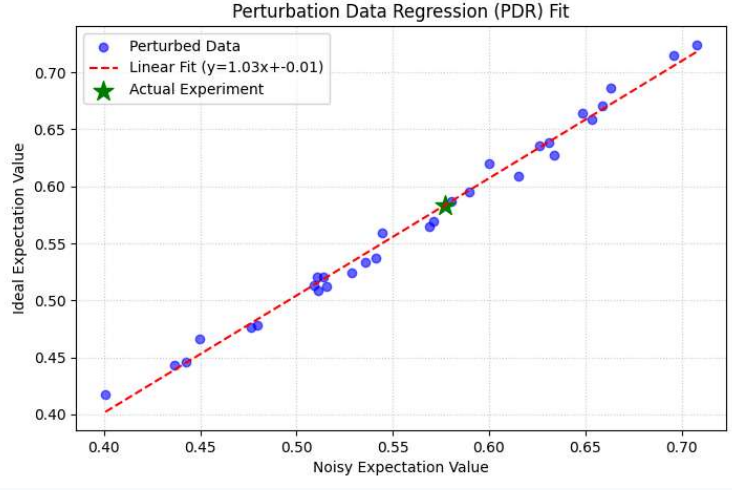


Figure 3.4: *Linear regression fit showing Noisy vs Ideal mapping.*

Experimental Setup

Component	Details
Simulated Backend	FakeFez (Noisy simulator of IBM Fez)
Real Hardware	IBM Fez (Superconducting Transmon)
Metric	Expectation Value ⟨ZZ⟩
Trials	100 statistical trials
Calibration Samples (K)	50 samples per variant

Results: Simulated (FakeFez)

Key Findings:

- **Linearity:** Median $R^2 > 0.90$ across all variants (Linear assumption holds).
- **Error Reduction:**
  - Unmitigated Error: 0.0184
  - Mitigated Error: **0.0014**
  - **92.5% Reduction**
- **Statistical Significance:** 95% CI includes the true ideal value ($p < 0.05$).

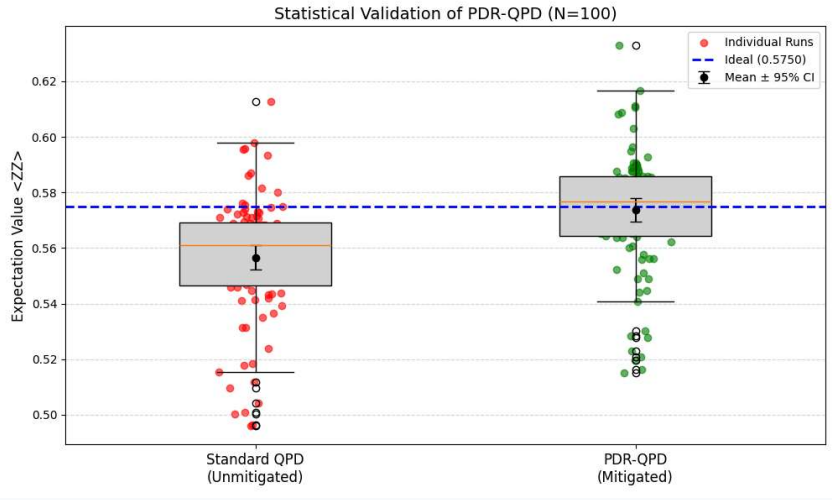


Figure 4.6: *Distribution of expectation values before and after mitigation.*

Results: Real Hardware (IBM Fez)

Performance on IBM Fez:

Method	Value ⟨ZZ⟩	Abs. Error
True Ideal	0.57502	-
Standard QPD	0.55563	0.01939
PDR-QPD	**0.56621**	**0.00881**

≈ 54.5% Error Reduction

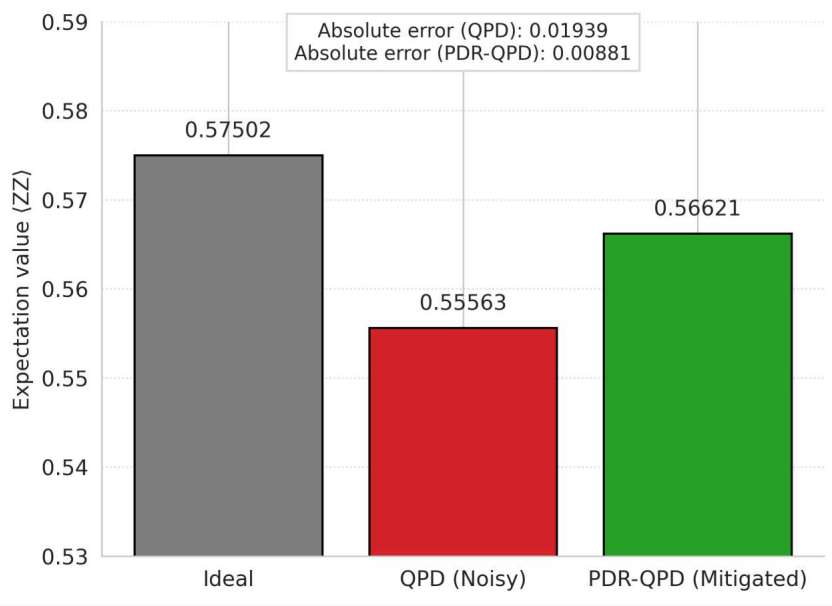


Figure 4.7: Comparison of Standard QPD vs PDR-QPD on real hardware.

Interpretation & Discussion

- **Validity of Linearity:** The localized linear model works effectively across different QPD variants, even on real hardware.
- **Adaptability:** The model successfully adapted to different noise regimes (Depolarizing in simulation vs. Coherent over-rotation on IBM Fez).
- **Cost/Benefit Analysis:**
  - Benefit: Significant reduction in bias (Systematic error).
  - Cost: QPD sampling overhead remains (exponential with cuts), plus linear calibration overhead.

Limitations & Future Work

Current Limitations

- **Sampling Overhead:** Scales exponentially with the number of cuts.
- **Linear Assumption:** May break down under strong non-linear noise regimes.
- **Calibration Cost:** Requires K runs for every circuit fragment.

Future Directions

- **Advanced Models:** Explore Gaussian Process Regression (GPR) or Bayesian models for non-linear noise.
- **Dynamic Cutting:** Optimize cut locations to minimize overhead.
- **Applications:** Test on VQE and QAOA algorithms for larger instances.

Conclusion



- **PDR-QPD Framework:** Combines structure-aware calibration with local linear correction and QPD reconstruction.
- **Validation:** Successfully validated on both FakeFez simulator and IBM Fez quantum processor.
- **Impact:** Achieved **54.5%** error reduction on hardware, proving that localized linear learning is a viable path for NISQ error mitigation.

Acknowledgements & Questions

Acknowledgements

I would like to thank my supervisor, **Dr. Rahat Hossain Faisal**, for their guidance, and the Department of CSE, University of Barisal.

Thank You!

Questions?

Appendix: Regression Statistics

Table 4.1: Statistical summary of regression parameters (100 trials).

Variant	Slope (m)	Intercept (c)	R^2 Score
Variant 1	0.9566 ± 0.0420	0.0409 ± 0.0244	0.9063 ± 0.0431
Variant 2	0.9542 ± 0.0463	0.0428 ± 0.0270	0.8986 ± 0.0451
Variant 3	0.9463 ± 0.0459	0.0468 ± 0.0269	0.9055 ± 0.0396
Variant 4	0.9426 ± 0.0432	0.0490 ± 0.0255	0.9010 ± 0.0478